Practical no. 9A

Aim: Create a PWA with a Web App Manifest file including:
App name
Icons
Start URL
Display mode
Implement Push Notifications in a PWA.

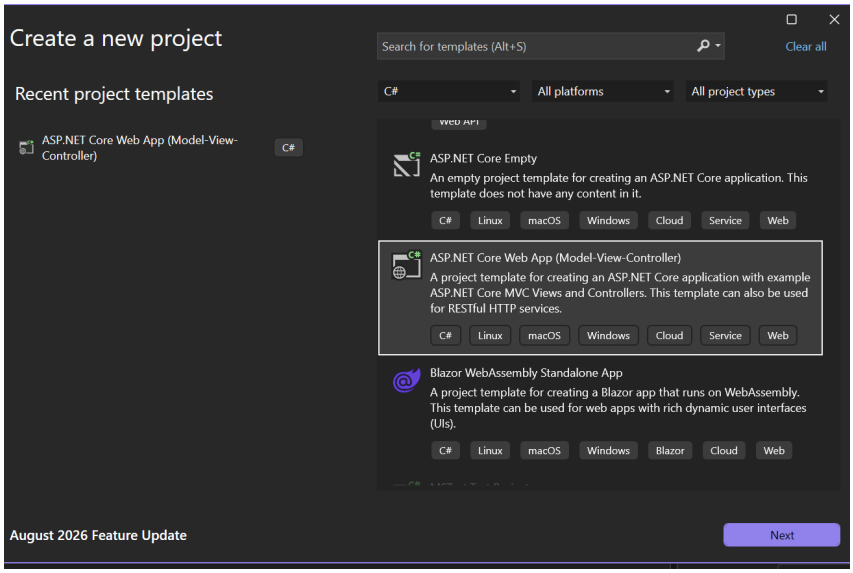
Software Required:

Visual Studio 2026

Procedure:

STEP 1: CREATE NEW PROJECT

1. Open Visual Studio 2026.
2. Click: Create a New Project
3. Open Visual Studio 2026.
4. Select Create a new project.
5. Search for ASP.NET Core Web App (Model-View-Controller).
6. Click Next.
7. Set the Project Name to Practical No. 9A.
8. Select a suitable location and click Next.
9. Select the installed/latest .NET version and keep the default options.
10. Click Create.



STEP 2: CREATE HOME PAGE

Open Views → Home → Index.cshtml and replace its contents with:

```
@{
    ViewData["Title"] = "PWA Home";
}

<div class="text-center">

    <h1>TYIT Student PWA</h1>

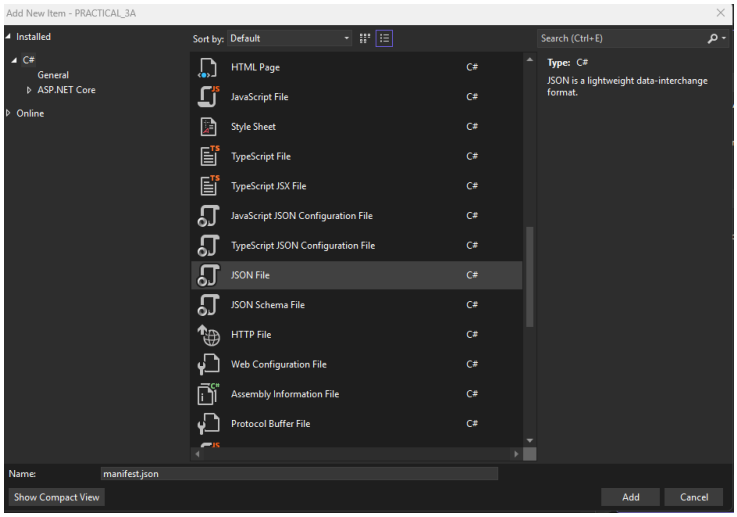
    <p>Progressive Web Application</p>

    <button id="notifyButton" class="btn btn-primary">
        Enable Notification
    </button>

</div>
<script src="/js/site.js"></script>
```

Step 5: Create Manifest File

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JSON file.
4. Enter File Name: **manifest.json**
5. Click Add.



Write the following code in the manifest.json file.

manifest.json

```
{
 "name": "TYIT Student PWA",
 "short_name": "TYIT PWA",
 "start_url": "/",
 "display": "standalone",
 "background_color": "#ffffff",
 "theme_color": "#0d6efd",
 "icons": [
  {
   "src": "/icons/sdsm.png",
   "sizes": "192x192",
   "type": "image/png"
  },
  {
   "src": "/icons/sdsm.png",
   "sizes": "512x512",
   "type": "image/png"
  }
 ]
}
```

Step 6: Add App Icon

1. Right Click → **wwwroot** folder
2. Click Add → New Folder
3. Enter Folder Name: **icons**
4. Paste one image in the **icons** folder.

Step 7: Edit _Layout.cshtml
Open:

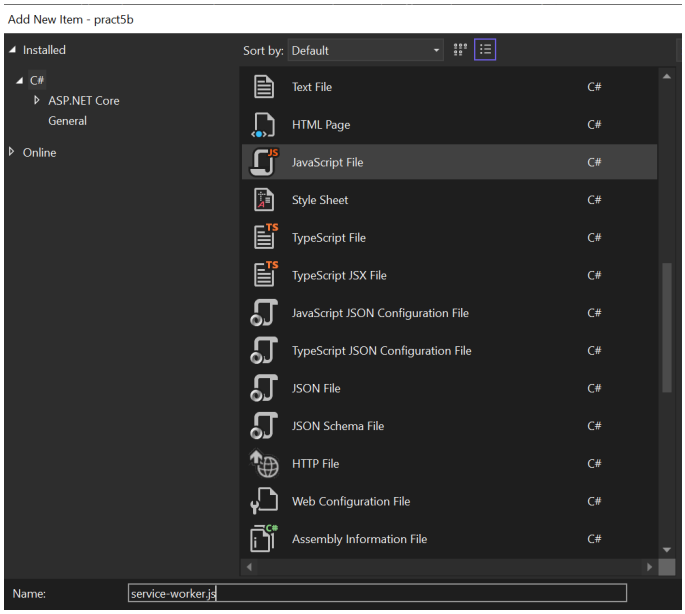
Views → Shared → _Layout.cshtml

Inside the <head> section, add:

```
<link rel="manifest" href="/manifest.json">
```

Step 8: Create Service Worker

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JavaScript file.
4. Enter File Name: **service-worker.js**
5. Click Add.



Write the following code in the service-worker.js file.

service-worker.js

```
self.addEventListener("install", function (event) {
    console.log("Service Worker Installed");
});

self.addEventListener("activate", function (event) {
    console.log("Service Worker Activated");
});
```

Step 9: Register the Service Worker
Open:

wwwroot → js → site.js

Add:

```
// Register Service Worker

if ("serviceWorker" in navigator) {

    window.addEventListener("load", function () {

        navigator.serviceWorker.register("/service-worker.js")
            .then(function (registration) {
```

```
            console.log("Service Worker Registered");

        })
        .catch(function (error) {

            console.log("Service Worker Registration Failed", error);

        });

    });

}

// Notification Button

document.getElementById("notifyButton").addEventListener("click", function () {

    if ("Notification" in window) {

        Notification.requestPermission().then(function (permission) {

            if (permission === "granted") {

                new Notification("TYIT Student PWA", {
                    body: "Welcome! Notification is working successfully.",
                    icon: "/icons/sdsm.png"
                });

            }

        });

    } else {

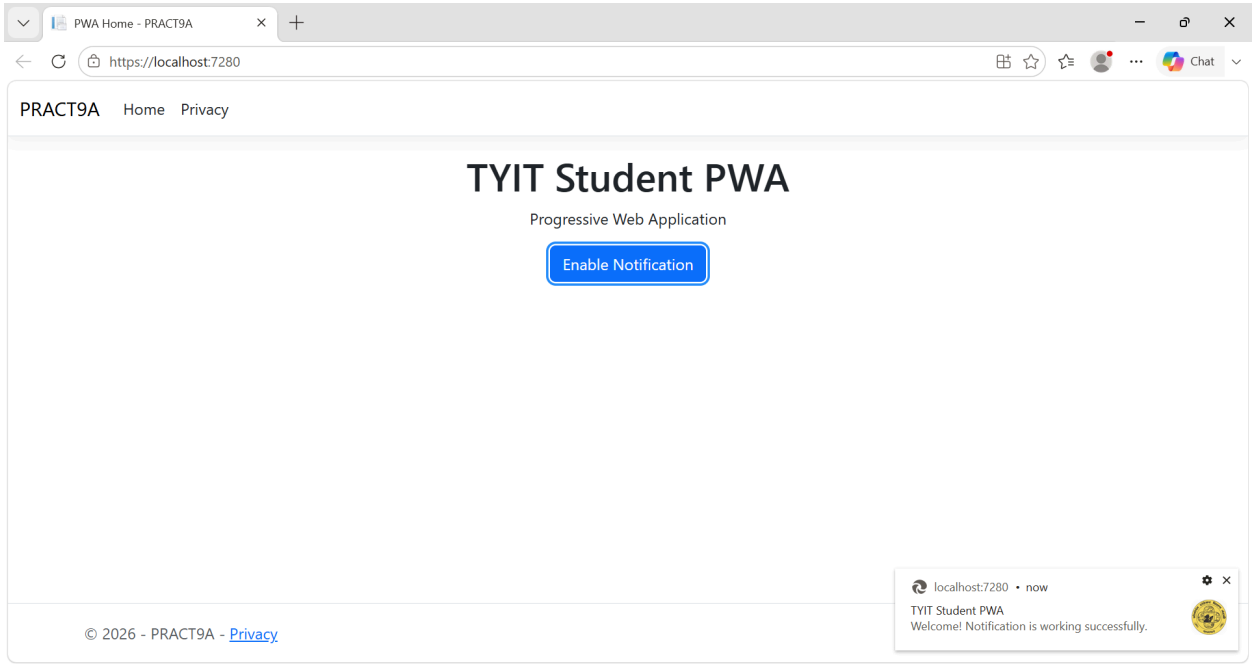
        alert("Notifications are not supported by this browser.");

    }

});
```

Step 10: Run the application
Press Ctrl + F5


PWA Home - pract9a
Get notifications?
localhost:5090
pract9a
http://localhost:5090 wants to
Show notifications
Allow
Block
TYIT Student PWA
Progressive Web Application
Enable Notification


PWA Home - PRACT9A
https://localhost:7280
Chat
PRACT9A
Home
Privacy
TYIT Student PWA
Progressive Web Application
Enable Notification
localhost:7280 • now
TYIT Student PWA
Welcome! Notification is working successfully.
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